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To the Editor-in-Chief  
*Statistics and Computing*

Dear Editor,

Please find enclosed our manuscript, “Anytime-Valid Nonparametric Testing of Synergistic (Group) Causality, with an Application Establishing Its Absence in Daily Financial Markets,” which we submit for consideration as an original research article in *Statistics and Computing*.

The paper addresses a question that recurs across the sciences but has lacked a principled sequential test: does a group of causes act on an effect jointly, beyond the sum of its members’ individual contributions, and can that be decided as data arrive under continuous monitoring? Every existing method for detecting such synergy—sufficient-cause interaction indices in epidemiology, joint-effect estimators in statistics, and information-theoretic decompositions such as PID, SURD, and PEID—is fixed-sample, returning a point value at a pre-committed sample size with no valid way to monitor evidence continuously. We close this gap by bringing the machinery of testing-by-betting and e-processes to bear on a super-additive synergy null.

Our main contributions are:

1. **ANTE**, an anytime-valid nonparametric test of synergy. A bounded, mean-centred interaction score is accumulated into a nonnegative capital process (e-process); Ville’s inequality delivers time-uniform Type-I error control under optional stopping and continuous monitoring, with no stationarity or correct-specification assumptions.
2. A theory of both validity and *power*: we prove the martingale (validity) property and its Ville corollary, and a growth/consistency theorem giving power one with an explicit detection-time bound under the alternative.
3. Extensions to  $k$ -way groups and to an out-of-sample predictive (Granger) synergy for nonstationary time series, together with a guarantee that ANTE isolates genuine interaction from mere joint dependence and within-variable nonlinearity—a discrimination competing estimators do not provide.
4. A substantive empirical finding: used as a demonstrably powered instrument, ANTE establishes (as evidence of absence, not absence of evidence) that super-additive group causality is absent from daily financial markets across thousands of asset triples, while the same test detects the turbulent energy cascade in physics and synthetic synergy at high power.

The work fits the scope of *Statistics and Computing*: a computational, sequential inference method with proven statistical guarantees, an efficient online algorithm, a head-to-head benchmark against 2024–2026 methods, and a fully reproducible real-data study with publicly available code and data (the repository link is included in the manuscript).

This manuscript is original, has not been published previously, and is not under consideration elsewhere. There are no competing interests, and no funding was received. The single author has approved the submission. We look forward to the reviewers’ feedback.

Sincerely,

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